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**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE**

**EXAMINATION NOVEMBER 2025**

**FIRST SEMESTER**

**Multi-Disciplinary Courses (MDC) - MG1MDCMBG100 - FASCINATING**

**WORLD OF MICROBES**

**(2024 ADMISSION ONWARDS)**

**Duration: 1.5 Hours** **Maximum Marks: 35**

**Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)**

**Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.**

**Part A**

**Short Answer Type Questions**

**Answer any 10 out of 12 Questions**

**Each Question carries 1 mark**

1. **Mention the important discovery of Alexander Fleming. [K] / [CO1]**
   Discovery of first antibiotic Penicillin
2. **Who is known as the Father of Microbiology ? [K] / [CO1]**
   Anton van Leeuwenhoek
3. **Which organism is used to produce Streptomycin? [K] / [CO2]**
   *Streptomyces griseus* of genus Streptomyces
4. **Who coined the term 'vaccine'? [K] / [CO1]**
   Edward jenner
5. **Name two bacteria which cause gas production in milk. [K] / [CO2]**
   *E.coli, Clostridium, Bacillus*

**6. State the theory of biogenesis. [K] / [CO1]**

Theory of biogenesis states that living organism can only originate from living organisms.

6. **What are Prokaryotes? [K] / [CO1]**
   Prokaryotes are single celled organisms that lack true nucleus and other membrane bound organelles.
7. **Name any two infections caused by Candida species. [K] / [CO2]**
   Oral candidiasis and Vulvovaginal candidiasis
8. **Define fermentation. [K] / [CO2]**
   Fermentation is a metabolic process where microorganisms convert sugars into acids, gases, or alcohol in the absence of oxygen
9. **Name one viral pesticide. [K] / [CO2]**
   NPV (Nuclear Polyhedro Virus, Baculovirus.
10. **What are the signs of spoiled fish? [K] / [CO2]**

The bright characteristic colors of the fish fade, and dirty, yellow, or brown discolorations appear.
The slime on the skin of the fish increases, especially at the flaps and gills.
The eyes gradually sink and shrink, the pupil becoming cloudy and the cornea opaque.
The gills turn a light pink and finally grayish-yellow color.

11. **Mention the selective medium for Mycobacteria. [K] / [CO2]**
    Lowenstein–Jensen medium or LJ medium

[10x1 = 10]

**Part B**

**Short Essay Type Questions**

**Answer any 3 out of 6 Questions**

**Each Question carries 5 marks**

12. **Write briefly on different types of Microorganisms with examples. [K] / [CO1]**
    Prokaryotes and Eukaryotes.(1)
    Microorganisms different types: bacteria, archaea, protozoa, algae, fungi, viruses, and Helminths.Explain briefly(4)
13. **State the characteristics of Kingdom Animalia. [K] / [CO1]**
    Could move
    Had a compact body appearance
    Heterotrophic nutrition
    Do not contain permanent vacuoles
    Showed quick response towards external stimuli
    Stored food in the form of glycogen and oil droplets
    Lack cellulosic cell wall.

Write any 5 points. **1 mark each**

14. **Explain the source, transmission and control of human corona virus. [E] / [CO2]**
    **Source -** an infected person's ,when they cough, sneeze, speak, sing or breathe. Caused by Covid virus.(1)
    **Transmission-** The virus is transmitted through direct contact with respiratory droplets of an infected person (generated through coughing and sneezing), and touching surfaces contaminated with the virus. The COVID-19 virus may survive on surfaces for several hours, but simple disinfectants can kill it.(2)
    **Control -** The best way to prevent and slow down transmission is to be well informed about the disease and how the virus spreads. Protect yourself and others from infection by staying at least 1 metre apart from others, wearing a properly fitted mask, and washing your hands or using an alcohol-based rub frequently. Get vaccinated when it's your turn and follow local guidance.(2)
15. **Describe the normal flora present on skin, mouth and intestine. [K] / [CO2]**
    The normal flora of the human body is the collection of microbes that live in a healthy person's body, and is also known as the resident flora or microbiota.(1)

Normal microflora of human skin are: (1.5)

- *Staphylococcus aureus*
- *Staphylococcus epidermidis*
- *Corynebacterium spp*
- *Propionibacterium spp*
- *Micrococcus spp*

Normal microflora of human mouth(1)

- In the mouth, yeast (Candida species) is present. The two protozoans most commonly found in the human mouth are *Entamoeba gingivalis and Trichomonas tenax.* Even in teeth, several anaerobes such as *Provotella spp, Fusobacterium spp, Lactobacilli and several anaerobic vibrios are seen.*

Normal microflora of human intestine(1.5)

The normal flora of the intestine, also known as the gut microbiome, is made up of a variety of bacteria, fungi, archaea, and viruses. Give examples.

16. **Mention five harmful microbes and their effects. [K] / [CO2]**
    Any five harmful microorganism and their effect. Eah carry 1 marks

- Eg E.coli - Urinary tract infection, Diarrhea, Septicemia, neonatal sepsis and meningitis and Pyogenic infections.
- Nipah
- Corona

**17. Why is Louis Pasteur considered as the father of Modern Microbiology? [K] / [CO1]**

Discuss the contributions of Louis Pasteur
Disproving Spontaneous Generation
Development of vaccines
Pasteurization
Fermentation

[3x5 = 15]

**Part C**

**Essay Type Questions**

**Answer any 1 out of 2 Question**

**Each Question carries 10 marks**

**19. State Koch's postulates. Which experiment led to the development of the postulates?[K] / [CO1]**

Koch's postulate (5)

Explain experiment with mice (5)

**20. Elaborate on beneficial microorganisms and their products. [C] / [CO2]**

**[1x10 = 10]**

Biofertilisers (2.5)

Biopesticides(2.5)

Fermentation industry (2.5)

Microorganism in pharmaceutical industry(2.5)